UBECON

Moving forward with event classification

Using machine learning to classify real cable events, with verified examples and expert review.

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A clearer view of cable behaviour.

Better context. Structured review.
A foundation for learning.

OUR FOCUS

Classify real cable events

Bring recordings, weather and expert knowledge together.

Use confirmed examples to teach the model and test its suggestions.

The platform is ready for structured investigation. Verified examples are the next priority.

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The building blocks for event classification are in place

The next phase brings these capabilities together around real recordings and confirmed examples.

CAPABILITY	WHAT IT ENABLES
Unified workspace	Explore recordings and retain expert conclusions
Mapped assets	Investigate cable behaviour across four routes
17 weather locations	Add temperature, wind and humidity context
Discovery workflow	Find useful examples for expert review
Machine learning	Random Forest classification trained on verified examples

Our focus is to turn real recordings and expert knowledge into dependable event classifications.

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A clear workflow for reviewing real recordings

The local workspace supports investigation, expert review and a lasting record of the evidence.

01 / EXPLORE

Choose a real recording

Find the relevant asset and inspect the recorded cable behaviour.

02 / UNDERSTAND

Bring the evidence together

Compare signal patterns with weather, inspections and confirmed observations.

03 / RECORD

Save the reviewed category

Keep the conclusion, reviewer and supporting evidence for learning and follow-up.

RECORDING + CONTEXT + EXPERT REVIEW

Every training example should connect a real recording to an evidence-backed label.

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The new UI brings map, signals and review together

A single workspace helps reviewers move from cable behaviour to a documented conclusion.



1. Explore the network

Search masts and routes. Use the map to choose where to investigate.

2. Inspect the recording

Review signal patterns and play back the recording to see changes over time.

3. Record the conclusion

Select an event to save its category, reviewer and supporting evidence.

UI DEMONSTRATION

Visible event entries are illustrative, not real incidents.

SCREENSHOT / local application; demonstration entries shown for interface illustration only

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Weather context now follows more of the network

Forecast coverage has expanded, and relative humidity is now being collected.

BEFORE

2 forecast locations

NOW

17 locations across the route network

COLLECTED

Temperature

COLLECTED

Wind speed

COLLECTED

Wind direction

NEW

Humidity

New records also retain when the forecast applies and when it was issued.

Richer weather context supports event interpretation; forecasts do not confirm the cause of an event.

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We use Random Forest machine learning for classification

Supervised learning: confirmed examples teach the model to recognise the four event categories.

THE CLUES

Cable behaviour + weather

How strong is the vibration?
Is it rhythmic or irregular?
What were the conditions?

MACHINE LEARNING TECHNIQUE

Random Forest

300 decision trees learn from confirmed examples. Their combined votes suggest the event category.

THE RESULT

A suggested category for review

The output includes a class and scores. Uncertain cases are flagged for attention.

INTENDED OUTPUTS

Normal operations

Ice loading

Wind galloping

External interference

The software can learn and make predictions; its reliability must be demonstrated on independent events.

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Two machine-learning steps support event classification

Unsupervised learning finds patterns; supervised learning links those patterns to confirmed event types.

01 / PILOT COMPLETED

Find patterns

K-means groups similar signals. Isolation Forest highlights unusual ones.

02 / NEXT PRIORITY

Confirm the cause

People use inspections or incident evidence to assign reliable labels.

03 / SOFTWARE READY

Teach the model

Random Forest learns the four event categories from expert-labeled examples.

04 / BEFORE ROLLOUT

Test new events

Check its suggestions on events it has never seen during training.

PILOT EVIDENCE

Discovery exercised on about 15 minutes of real recordings.

Finding an unusual pattern is a way to select examples for review; it does not establish an event class.

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The goal is to distinguish four types of operation

Each category needs confirmed examples, including cases that are easy to confuse.

01

Normal operations

Recognise normal cable behaviour, including windy conditions.

02

Ice loading

Test for changes associated with independently confirmed ice.

03

Wind galloping

Recognise sustained cable movement confirmed in the field.

04

External interference

Investigate disturbances such as strikes or vegetation contact.

A quiet signal is not automatically normal, and cold or humid weather is not proof of ice.

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Collect confirmed examples of real cable behaviour

Build a dedicated training dataset from verified recordings and independent field evidence.

Real data

Verified labels

A dedicated dataset for training and testing the classifier.

01 **Match the evidence**

Confirm when and where a recording belongs.

02 **Verify the cause**

Use inspections, cameras or reliable incident reports.

03 **Include normal operation**

Teach the model what ordinary conditions look like.

The most valuable next asset is a small, trustworthy set of examples that can grow over time.

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Start with verified examples, then classify with oversight

A short sequence takes the project from the current platform to useful event suggestions.

STEP 1

Select and align

Choose pilot recordings and verify time, place and weather.

STEP 2

Confirm the labels

Have experts review examples of all four categories.

STEP 3

Train and test

Measure correct classifications, missed events and false alerts.

STEP 4

Use with review

Pilot model suggestions while people check the results.

FIRST MILESTONE: an expert-reviewed dataset with evidence for every label.

Management focus: agree the pilot scope and assign a field-review owner to build the first labeled dataset.